


# **L11a**: Composite Contracts and Their Probability of Profit

CHEME 5660 Cornell University

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# Objectives for Today

Today, we combine single contracts into composite contracts. What is a composite contract worth, and when does it make money?

**Today's concepts**

- The NPV and POP of a composite contract: sums over its legs.
- Common composite contracts: outlook, breakevens, maximum profit and loss.
- Closing before expiration: why an early exit earns less.

# What You Need to Know

## Core concepts

- Profit at expiration: the legs' payoffs, weighted by $\theta_i n_i$, minus the composite premium grown to expiration. POP: the probability it is positive.
- Read a profit diagram: the outlook, the breakevens, the maximum profit (the credit grown to expiration), and the maximum loss.
- Closing early earns less, by roughly the extrinsic value left in the legs. Near expiration, a small move near a strike changes the profit by more.

## **For understanding** (you won't reproduce these)

- The NPV derivation and the formula tables for the eight composite contracts.

# Lectures and Examples

**Lecture:**
CHEME-5660-L11a-Lecture-CompositeContracts-Fall-2026.ipynb

**Example 1:**
CHEME-5660-L11a-Example-Delta-VerticalSpread-Fall-2026.ipynb

**Example 2:**
CHEME-5660-L11a-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

# Composite Contracts

A composite contract $\mathcal{C}$ combines long and short puts and calls on one underlying asset with one expiration date $T$. Its payoff per share at expiration is:

$$\hat{V}_{\mathcal{C}}(S(T)) = \underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, \overbrace{V_i(K_i, S(T))}^{\text{payoff of leg } i}}_{\text{sum of all legs}}$$

Here $S(T)$ is the share price at expiration, $K_i$ the strike price of leg $i$, $V_i$ its payoff, $(S(T) - K_i)^+$ for a call or $(K_i - S(T))^+$ for a put, $n_i$ its number of contracts, and $\theta_i = -1$ if short or $1$ if long.

# The Net Present Value of a Composite Contract

A composite contract is a sum of legs, so its NPV is the sum of the legs' NPVs, weighted by $\theta_i n_i$:

$$\mathrm{NPV}_0(\mathcal{C}) = \sum_{i \in \mathcal{C}} \theta_i \, n_i \underbrace{\Big( -\mathcal{P}_i + \overbrace{\mathcal{D}_{T,0}^{-1}(g_y) \cdot V_i(K_i, S(T))}^{\text{discounted payoff}} \Big)}_{\text{NPV of buying leg } i} \qquad \text{weighted leg NPVs}$$

$$= -\underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, \mathcal{P}_i}_{= \mathcal{P}_{\mathcal{C}}} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, V_i(K_i, S(T))}_{= \hat{V}_{\mathcal{C}}(S(T))} \qquad \text{split the sum}$$

$$= -\mathcal{P}_{\mathcal{C}} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot \hat{V}_{\mathcal{C}}(S(T))$$

where $\mathcal{P}_i$ is the premium of leg $i$ (USD/share), positive whether we buy or sell it ($\theta_i$ sets the direction), $\mathcal{P}_{\mathcal{C}}$ is the composite premium, and $\hat{V}_{\mathcal{C}}(S(T))$ is the composite payoff.

# Profit at Expiration

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ writes the NPV as a profit at expiration discounted to today:

$$\mathrm{NPV}_0(\mathcal{C}) = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\left(\hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}\right)}_{= \hat{P}_{\mathcal{C}}(S(T))} \qquad \text{factor out } \mathcal{D}_{T,0}^{-1}(g_y)$$

$$= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\hat{P}_{\mathcal{C}}(S(T))}_{\text{profit at expiration}}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ is the accumulation factor, and:

$$\boxed{\hat{P}_{\mathcal{C}}(S(T)) = \hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}}$$

is the profit at expiration (USD/share).

# Debit or Credit

Because $\theta_i = 1$ for a long leg and $\theta_i = -1$ for a short leg, the composite premium splits into the premium we pay and the premium we collect:

$$\mathcal{P}_{\mathcal{C}} = \underbrace{\sum_{\text{long legs}} n_i \mathcal{P}_i}_{\text{premium we pay}} - \underbrace{\sum_{\text{short legs}} n_i \mathcal{P}_i}_{\text{premium we collect}}$$

If we pay more than we collect, $\mathcal{P}_{\mathcal{C}} > 0$: a debit. If we collect more than we pay, $\mathcal{P}_{\mathcal{C}} < 0$: the contract has a negative price to a buyer, so we are paid to open it, a credit of $-\mathcal{P}_{\mathcal{C}}$ USD/share. For example, selling one put with premium $\mathcal{P}_p$ gives $\mathcal{P}_{\mathcal{C}} = -\mathcal{P}_p$, a credit of $\mathcal{P}_p$.

# Probability of Profit and Probability of Exercise

The NPV and the profit at expiration have the same sign, so with $\mathbb{P}$ our belief, the probability of profit is:

$$\boxed{\text{POP}_{\mathcal{C}} = \mathbb{P}\Big(\text{NPV}_0(\mathcal{C}) > 0\Big) = \mathbb{P}\Big(\hat{P}_{\mathcal{C}}(S(T)) > 0\Big)}$$

Each leg has its own probability of exercise:

$$\text{POE}_i = \begin{cases} \mathbb{P}\left(S(T) > K_i + 0.01\right) & \text{if leg } i \text{ is a call} \\ \mathbb{P}\left(S(T) < K_i - 0.01\right) & \text{if leg } i \text{ is a put} \end{cases}$$

# Composite Credit Contracts: Outlook

The eight structures, by side:

| Composite contract | Put side | Call side | Outlook |
|---|---|---|---|
| Short put | short put at $K$ | none | neutral to bullish |
| Short call | none | short call at $K$ | neutral to bearish |
| Put credit spread | short put at $K_2$<br>long put at $K_1 < K_2$ | none | neutral to bullish |
| Call credit spread | none | short call at $K_1$<br>long call at $K_2 > K_1$ | neutral to bearish |
| Short straddle | short put at $K$ | short call at $K$ | neutral |
| Short strangle | short put at $K_1$ | short call at $K_2 > K_1$ | neutral |
| Protected strangle | short put at $K_1$ | short call at $K_2$<br>long call at $K_3 > K_2$ | neutral, leaning bullish |
| Iron condor | short put at $K_2$<br>long put at $K_1 < K_2$ | short call at $K_3$<br>long call at $K_4 > K_3$ | neutral |

Strikes are numbered low to high, $\mathcal{P}_i$ is the premium of the leg at $K_i$. Figures: payoff (blue), profit (black), and $S(0)$, the share price at entry.

# Composite Credit Contracts: Three Steps

Each row of the tables follows from the profit at expiration expression in three steps:

1. Compute the composite premium $\mathcal{P}_{\mathcal{C}}$, with a minus sign for each short leg. If $\mathcal{P}_{\mathcal{C}} < 0$, the contract opens for a credit of $-\mathcal{P}_{\mathcal{C}}$ (a positive term in the profit at expiration).
2. Write the profit on each interval between the strikes. There, each leg's payoff is zero or linear in $S(T)$, so the profit is a straight line.
3. Evaluate the profit at the strikes, at $S(T) = 0$, and as $S(T)$ grows, to find the maximum profit and maximum loss. Set the profit to zero where it slopes to find the breakevens, keeping each solution that lies in its interval.

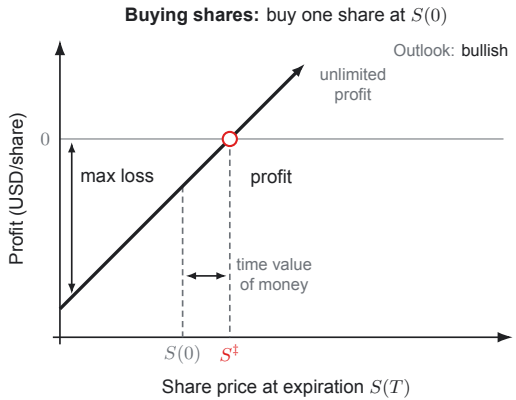
Maximum losses are listed as positive amounts (USD/share).

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**Derivation:** CHEME-5660-L11a-Derivation-ProfitLossBreakeven-Fall-2026.ipynb

# Buying Shares

Before combining options, let's look at the simplest position: buying one share at $S(0)$ and holding it to expiration:



# Buying Shares: Profit at Expiration

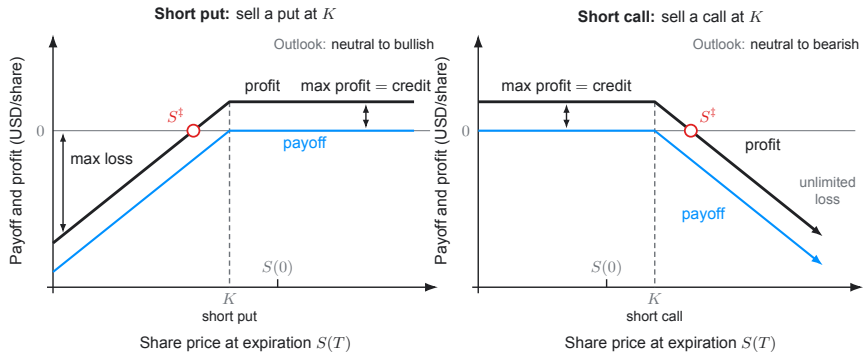
The profit at expiration is the share price at expiration minus the purchase price grown to expiration:

$$\hat{P}(S(T)) = S(T) - \mathcal{D}_{T,0}(g_y) \cdot S(0)$$

It breaks even at $S^{\ddagger} = \mathcal{D}_{T,0}(g_y) \cdot S(0)$, slightly above $S(0)$, because the money paid for the share could have earned the risk-free rate. The profit has no upper limit, but if the share price falls to zero, we lose the whole purchase price grown to expiration.

# Short Puts and Short Calls

Selling one put, a short put, or one call, a short call, is the building block of every structure below:




Each keeps the whole premium if it expires out of the money.

# Short Puts and Short Calls: Profit at Expiration

With premiums $\mathcal{P}_p$ (put) and $\mathcal{P}_c$ (call) at strike $K$, the profits at expiration, and the breakevens $S^\ddagger$, are:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p - (K - S(T))^+ \qquad \text{short put}$$

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c - (S(T) - K)^+ \qquad \text{short call}$$

| | Short put | Short call |
|---|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ for $S(T) \geq K$ | $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$ for $S(T) \leq K$ |
| Breakeven | $S^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ | $S^\ddagger = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$ |
| Maximum loss | $K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ at $S(T) = 0$ | unlimited |

The short put's loss is bounded ($S(T) \geq 0$); the short call's is not.

# Short Put Compared with One Share

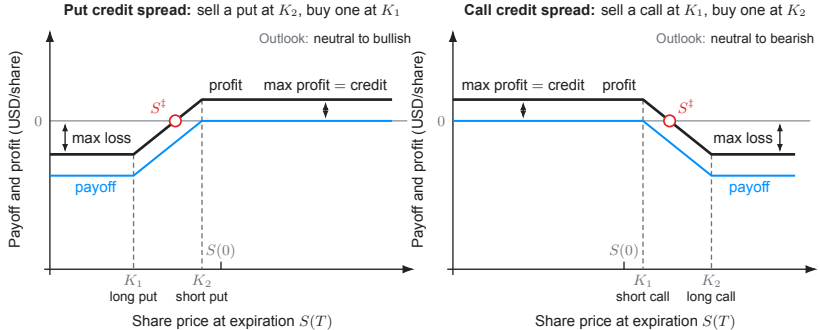
Next to one share bought at $S(0)$, the short put has:

| | Short put | One share |
|---|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ for $S(T) \geq K$ | unlimited |
| Breakeven | $S^{\ddagger} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ | $S^{\ddagger} = \mathcal{D}_{T,0}(g_y) \cdot S(0)$ |
| Maximum loss | $K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ at $S(T) = 0$ | $\mathcal{D}_{T,0}(g_y) \cdot S(0)$ at $S(T) = 0$ |

With $S(0) = K$, the short put breaks even lower and has a smaller maximum loss, but its profit stops at the credit grown to expiration. All entries are per share: one contract covers 100 shares, so multiply by 100 to compare one contract with 100 shares.

# Credit Spreads

A credit spread sells the contract closer to the share price and buys one farther away, same type and expiration (a bull put or bear call spread):




The short leg collects the credit; the long leg caps the loss.

# Put Credit Spread

A long put at $K_1$ and a short put at $K_2$: the credit grown to expiration, minus and plus the leg payoffs:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1) - (K_2 - S(T))^+ + (K_1 - S(T))^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \geq K_2$ |
| Breakeven | $S^\ddagger = K_2 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ |
| Maximum loss | $(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \leq K_1$ |

**Example 1:** CHEME-5660-L11a-Example-Delta-VerticalSpread-Fall-2026.ipynb

# Call Credit Spread

A short call at $K_1$ and a long call at $K_2$ give the profit at expiration:

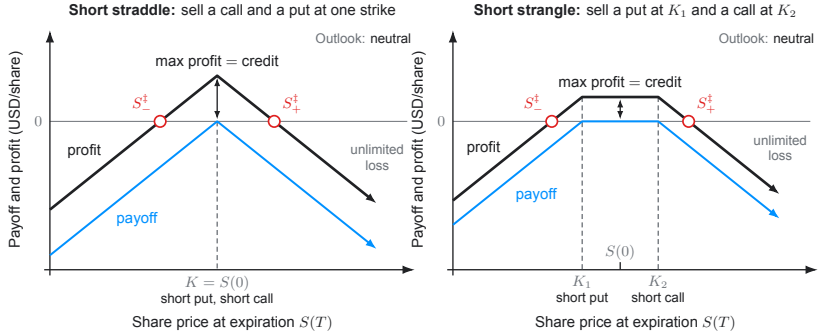
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2) - (S(T) - K_1)^+ + (S(T) - K_2)^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \leq K_1$ |
| Breakeven | $S^\ddagger = K_1 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ |
| Maximum loss | $(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \geq K_2$ |

Beyond the long strike, the long leg gains what the short leg loses, so the loss stops at the strike width minus the credit.

# Short Straddles and Strangles

A straddle: a put and a call at one strike $K$. A strangle: a put at $K_1$ below a call at $K_2$. Same expiration:




Sold, each profits only between its breakevens: POP $= \mathbb{P}(S_-^\ddagger < S(T) < S_+^\ddagger)$.

# Short Straddle

Selling a put and a call at the strike $K$ gives the profit at expiration:

$$\widehat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c) - (K - S(T))^+ - (S(T) - K)^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = K$ |
| Lower breakeven | $S_-^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ |
| Maximum loss below $S_-^\ddagger$ | $K - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = 0$ |
| Upper breakeven | $S_+^\ddagger = K + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ |
| Maximum loss above $S_+^\ddagger$ | unlimited |

The straddle collects more credit, but the strangle keeps its credit over a wider range of share prices.

# Short Strangle

Selling a put at $K_1$ and a call at $K_2$ gives the profit at expiration:

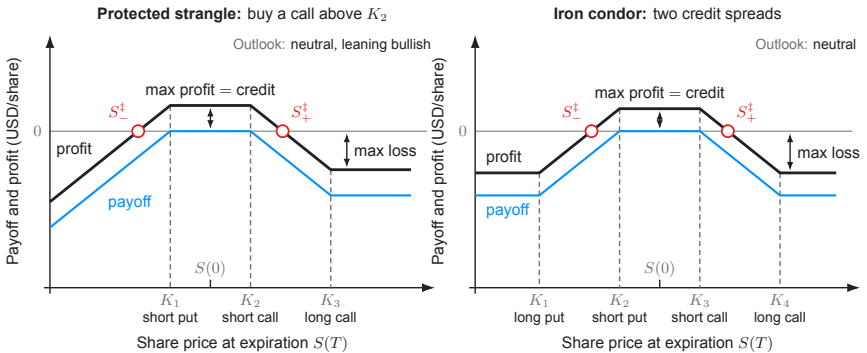
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2) - (K_1 - S(T))^+ - (S(T) - K_2)^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ for $K_1 \leq S(T) \leq K_2$ |
| Lower breakeven | $S_-^\ddagger = K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ |
| Maximum loss below $S_-^\ddagger$ | $K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ at $S(T) = 0$ |
| Upper breakeven | $S_+^\ddagger = K_2 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ |
| Maximum loss above $S_+^\ddagger$ | unlimited |

**Example 2:** CHEME-5660-L11a-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

# Protected Strangles and Iron Condors

A protected strangle (jade lizard) adds a long call at $K_3$ to a short strangle; an iron condor also adds a long put:




Each long leg caps the loss on its side but lowers the credit.

# Protected Strangle

A short put at $K_1$, a short call at $K_2$, and a long call at $K_3$ give the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) \qquad \text{protected strangle}$$
$$- (K_1 - S(T))^+ - (S(T) - K_2)^+ + (S(T) - K_3)^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ for $K_1 \leq S(T) \leq K_2$ |
| Lower breakeven | $S_-^\ddagger = K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ |
| Maximum loss below $S_-^\ddagger$ | $K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ at $S(T) = 0$ |
| Upper breakeven | $S_+^\ddagger = K_2 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)^a$ |
| Maximum loss above $S_+^\ddagger$ | $(K_3 - K_2) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ for $S(T) \geq K_3{}^b$ |

[a] If $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) \leq K_3 - K_2$; otherwise none. [b] If $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) < K_3 - K_2$; otherwise none.

If the credit grown to expiration is at least $K_3 - K_2$, the call side cannot lose.

# Iron Condor

A long put at $K_1$, a short put at $K_2$, a short call at $K_3$, and a long call at $K_4$ give the profit at expiration:

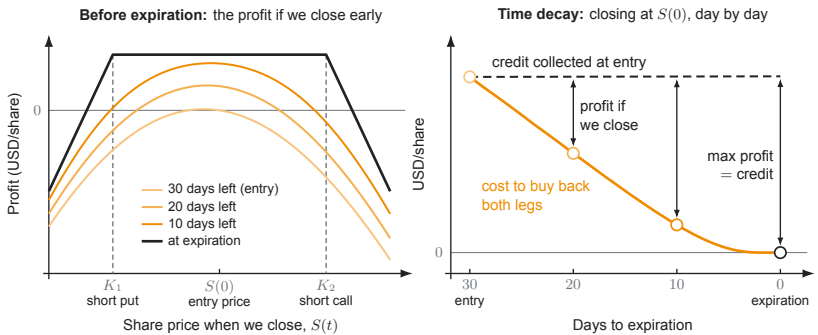
$$\begin{aligned}\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4) \\ + (K_1 - S(T))^+ - (K_2 - S(T))^+ - (S(T) - K_3)^+ + (S(T) - K_4)^+\end{aligned}$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $K_2 \le S(T) \le K_3$ |
| Lower breakeven | $S_-^\ddagger = K_2 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ |
| Maximum loss below $S_-^\ddagger$ | $(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \le K_1$ |
| Upper breakeven | $S_+^\ddagger = K_3 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ |
| Maximum loss above $S_+^\ddagger$ | $(K_4 - K_3) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \ge K_4$ |

At any share price at expiration, at most one side can lose, so the maximum loss is the wider strike width minus the credit.

# Why Wait to Expiration?

Our diagrams are drawn at expiration. Closing a short leg early means buying it back: intrinsic plus extrinsic value $E_V(t)$. A short strangle (put $K_1 = 90$, call $K_2 = 110$, $S(0) = 100$ USD/share, 30 days, CRR tree, $\sigma = 0.40$, $g_y = 4\%$):



# Why Wait to Expiration? What Do We See?

- Before expiration (left): the buyback price still includes extrinsic value, so earlier curves lie below the profit at expiration.
- At the entry share price (right): both legs are out of the money, so closing at once costs exactly the credit and earns nothing.
- Time decay: as the extrinsic value decays, the profit if we close at $S(0)$ rises to the whole credit.

The diagrams show the most each contract can earn; an early exit earns less, by roughly the extrinsic value left in the legs.

# Summary

## Key takeaways

- The NPV of a composite contract is a sum over its legs: the POP is the probability that the profit at expiration is positive.
- Common composite contracts can be bullish, bearish, or neutral: the profit diagram gives the outlook, breakevens, and maximum loss.
- Closing before expiration earns less: buying the legs back still pays for their extrinsic value.